Applications of Derivatives

File 1 of 3 · Detailed Teaching Notes

Covers the complete textbook chapter (pp. 65–94): tangents & normals, derivative as a rate measure, velocity–acceleration–jerk, approximations, Rolle's theorem, LMVT, increasing & decreasing functions, maxima & minima and optimisation problems. Every board question on this chapter from the six papers in your project (Mar 2024 – Jun 2026) is tagged and solved where it appears. Read this first; then use File 2 for revision and File 3 for timed practice.

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1 How to use this chapter (read this first)

1.1 What the chapter is really about

Chapter 1 taught you *how* to differentiate. Chapter 2 asks: **what can a derivative tell you?** It turns out one number, $f'(a)$, carries three different meanings, and the whole chapter is just these three meanings applied:

Meaning of $f'(a)$	Topic in this chapter	Typical question
Slope of the tangent at $x = a$	Tangents & normals; Rolle; LMVT	"Find the equation of the tangent at ..."
Rate of change of y w.r.t. x	Rate measure; velocity, acceleration	"How fast is the area increasing ..."
Local behaviour (going up / down / flat)	Approximation; increasing–decreasing; maxima–minima	"Find the dimensions for maximum volume"

If you remember that $f'(a)$ is simultaneously a slope, a rate and a direction signal, nothing in this chapter is new — it is the same tool pointed at different problems.

1.2 How much this chapter is worth (observed, not official)

The Board does not publish chapter-wise weightage. What follows is what I *counted* in the six papers in your project folder. Treat it as a pattern, not a promise.

Paper	Ch. 2 questions offered	Marks offered
Mar 2024 (J-862)	MCQ slope of tangent (2) · approx. $\tan^{-1}(1.002)$ (3) · open box max volume (4)	9
Jul 2024 (J-174)	MCQ approx. $\tan(44°30')$ (2) · tangent parallel to a line (2) · particle acceleration (2) · approx. of polynomial (3)	9
Feb 2025 (J-312)	VSA condition for increasing (1) · divide 20 into two parts (2) · velocity at given acceleration (3) · tangent & normal (3)	9
Jul 2025 (J-384)	MCQ acceleration (2) · tangent parallel to line (2) · Rolle conditions (2) · water in cylinder rate (3)	9
Feb 2026 (J-165)	MCQ approximation (2) · increasing test (2) · circular wave rate (2) · verify LMVT (3)	9
Jun 2026 (J-276)	VSA velocity (1) · increasing test (2) · wire-rectangle max area (3) · verify Rolle (3)	9

The single most useful observation

In all six papers, **every** Chapter 2 question was either a textbook solved example or a textbook exercise question, often word-for-word with changed numbers. The chapter is short (30 pages). Solving every example and exercise in pp. 65–94 is the highest-return preparation you can do for these 9 marks.

Notice also *where* the marks sit: mostly 2- and 3-mark questions, with one 4-mark optimisation question in Mar 2024. Optimisation (maxima/minima word problems) is the only topic long enough to be a 4-marker.

1.3 Prerequisites you must already have (from Chapter 1)

You will use these constantly. If any feels shaky, revise it before going on.

Chapter 1 tools used in Chapter 2

- **Standard derivatives:** $\frac{d}{dx}x^n = nx^{n-1}$, $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = \sec^2 x$, $(e^x)' = e^x$, $(\log x)' = \frac{1}{x}$, $(\tan^{-1} x)' = \frac{1}{1+x^2}$, $(\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$, $(\cos^{-1} x)' = -\frac{1}{\sqrt{1-x^2}}$.
- **Chain rule:** $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$. Related-rate problems are nothing but the chain rule with t (time).
- **Implicit differentiation:** differentiate both sides w.r.t. x, treating y as a function of x, so $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$.
- **Parametric:** $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$, provided $\frac{dx}{dt} \neq 0$.
- **Second derivative:** $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$.

Notation note: the textbook writes $\log x$ for the natural logarithm $\log_e x$. So $\frac{d}{dx}\log x = \frac{1}{x}$. When base 10 is intended, it writes $\log_{10} x$ explicitly.

2 Tangents and Normals (Textbook 2.1.2)

2.1 Intuition

Zoom in far enough on a smooth curve at a point P and the curve looks like a straight line. That line is the **tangent** at P. Its slope is the derivative at P — this is the geometrical meaning of the derivative you met in Chapter 1 (Section 1.2.1).

The **normal** at P is the line through P perpendicular to the tangent. Think of a ball rolling along a curved track: the tangent is the direction it moves in; the normal is the direction the track pushes on it.

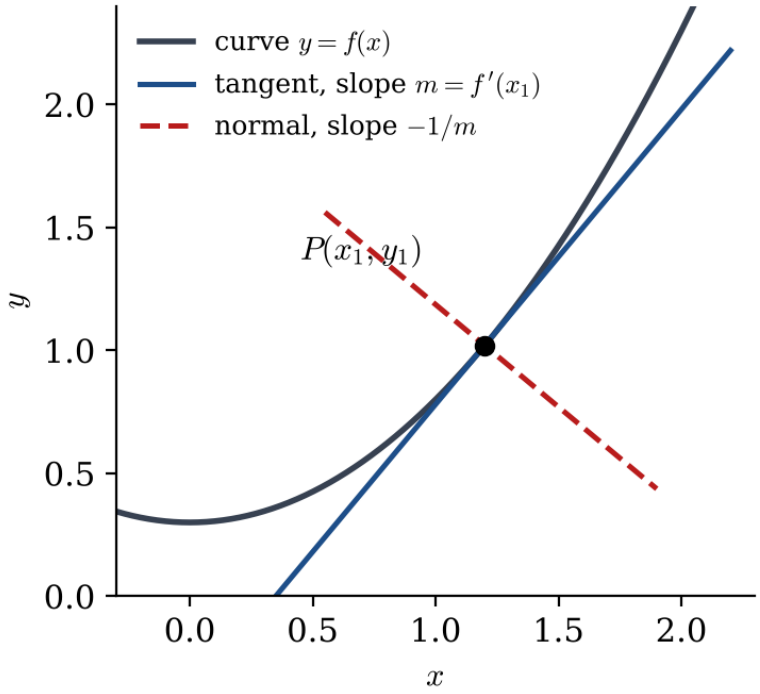


Figure 1: Tangent and normal at a point $P(x_1, y_1)$ on $y = f(x)$

2.2 The formulas and why they hold

Slopes and equations at $P(x_1, y_1)$

Slope of tangent:

$$m = \left(\frac{dy}{dx}\right)_{(x_1,y_1)}$$

Slope of normal (when $m \neq 0$):

$$m' = -\frac{1}{m} = -\frac{1}{\left(\frac{dy}{dx}\right)_{(x_1,y_1)}}$$

Equation of tangent: $y - y_1 = m(x - x_1)$
Equation of normal: $y - y_1 = m'(x - x_1)$

Why $m' = -1/m$? Two non-vertical lines with slopes m_1, m_2 are perpendicular exactly when $m_1 m_2 = -1$. The normal is perpendicular to the tangent, so $m \cdot m' = -1$, giving $m' = -1/m$.

Why the point–slope form? A line through (x_1, y_1) with slope m: for any other point (x, y) on it, $\dfrac{y - y_1}{x - x_1} = m$. Cross-multiplying gives $y - y_1 = m(x - x_1)$.

2.3 Special cases the formula hides

Situation at P	Tangent	Normal
$\dfrac{dy}{dx} = 0$	horizontal: $y = y_1$	vertical: $x = x_1$
$\dfrac{dy}{dx}$ undefined (denominator 0, e.g. $\dfrac{dx}{dy} = 0$)	vertical: $x = x_1$	horizontal: $y = y_1$
Tangent **parallel** to a line of slope k	set $\dfrac{dy}{dx} = k$	—
Tangent **perpendicular** to a line of slope k	set $\dfrac{dy}{dx} = -\dfrac{1}{k}$	—
Normal **parallel** to a line of slope k	set $-\dfrac{1}{dy/dx} = k$	—
Tangent parallel to X-axis	$\dfrac{dy}{dx} = 0$	
Tangent parallel to Y-axis	$\dfrac{dx}{dy} = 0$	

Slope of the line $ax + by + c = 0$ is $-\dfrac{a}{b}$. You will need this in almost every "parallel / perpendicular to a given line" question.

Mistakes that cost marks here

1. **Substituting the point too early.** Differentiate first, *then* substitute (x_1, y_1). If you substitute first you are differentiating a constant and get 0.
2. **Forgetting to find the point** in parametric questions. Given θ, you must compute *both* x and y before writing the equation.
3. **Sign slip in the normal.** Normal slope is $-1/m$, not $1/m$ and not $-m$.
4. **Leaving the answer as $y - 2 = \frac{1}{2}(x - \frac{1}{2})$.** Board answers are expected in the form $ax + by + c = 0$ with integer coefficients. Clear fractions.

2.4 Worked examples (textbook Ex. 1 and Ex. 2, p. 66–68)

Example 1 — explicit curve [PYQ Feb 2025, Q23, 3 marks]

Find the equations of the tangent and normal to $y = 2x^3 - x^2 + 2$ at $\left(\frac{1}{2}, 2\right)$.

Step 1 — differentiate.

$$\frac{dy}{dx} = 6x^2 - 2x$$

Step 2 — slope at the point.

$$m = \left(\frac{dy}{dx}\right)_{\left(\frac{1}{2},2\right)} = 6\left(\tfrac{1}{2}\right)^2 - 2\left(\tfrac{1}{2}\right) = \tfrac{3}{2} - 1 = \tfrac{1}{2}$$

So the slope of the normal is $m' = -\dfrac{1}{m} = -2$.

Step 3 — tangent.

$$y - 2 = \tfrac{1}{2}\left(x - \tfrac{1}{2}\right) \;\Rightarrow\; 4y - 8 = 2x - 1 \;\Rightarrow\; 2x - 4y + 7 = 0$$

Step 4 — normal.

$$y - 2 = -2\left(x - \tfrac{1}{2}\right) \;\Rightarrow\; y - 2 = -2x + 1 \;\Rightarrow\; 2x + y - 3 = 0$$

Check the point first: $2\left(\tfrac{1}{8}\right) - \tfrac{1}{4} + 2 = \tfrac{1}{4} - \tfrac{1}{4} + 2 = 2$. ✓ The point lies on the curve. Doing this check takes five seconds and catches copying errors in the question.

Example 2 — implicit curve

Find the tangent and normal to $x^3 + 2x^2y - 9xy = -2$ at $(2,1)$.
Differentiate both sides w.r.t. x. The term $2x^2y$ is a product, so use the product rule:

$$3x^2 + 2\left[x^2\frac{dy}{dx} + 2xy\right] - 9\left[x\frac{dy}{dx} + y\right] = 0$$

$$3x^2 + 2x^2\frac{dy}{dx} + 4xy - 9x\frac{dy}{dx} - 9y = 0$$

Collect the $\dfrac{dy}{dx}$ terms:

$$(2x^2 - 9x)\frac{dy}{dx} = 9y - 4xy - 3x^2 \quad\Rightarrow\quad \frac{dy}{dx} = \frac{9y - 4xy - 3x^2}{2x^2 - 9x}$$

At $(2,1)$:

$$m = \frac{9 - 8 - 12}{8 - 18} = \frac{-11}{-10} = \frac{11}{10}$$

A known error in the textbook

The textbook solution of this example (p. 66) writes the denominator as $2(4) - 9(1) = -1$ and gets $m = 11$. That is an arithmetic slip: the denominator is $2x^2 - 9x$ evaluated at $x = 2$, which is $8 - 18 = -10$. The correct slope is $\frac{11}{10}$, giving
Tangent: $y - 1 = \frac{11}{10}(x - 2) \;\Rightarrow\; 11x - 10y - 12 = 0$
Normal: $y - 1 = -\frac{10}{11}(x - 2) \;\Rightarrow\; 10x + 11y - 31 = 0$
If this exact question appears, write the correct working. Examiners mark the mathematics, not the textbook's typo.

Example 3 — parametric curve

Find the tangent and normal to $x = 2\sin^3\theta$, $y = 3\cos^3\theta$ at $\theta = \dfrac{\pi}{4}$.

$$\frac{dx}{d\theta} = 6\sin^2\theta\cos\theta, \qquad \frac{dy}{d\theta} = -9\cos^2\theta\sin\theta$$

$$\frac{dy}{dx} = \frac{-9\cos^2\theta\sin\theta}{6\sin^2\theta\cos\theta} = -\frac{3}{2}\cot\theta \quad \Rightarrow \quad m = -\frac{3}{2}\cot\frac{\pi}{4} = -\frac{3}{2}, \qquad m' = \frac{2}{3}$$

The point (do not skip): $x = 2\left(\frac{1}{\sqrt{2}}\right)^3 = \frac{1}{\sqrt{2}}$, $y = 3\left(\frac{1}{\sqrt{2}}\right)^3 = \frac{3}{2\sqrt{2}}$.
Tangent:

$$y - \frac{3}{2\sqrt{2}} = -\frac{3}{2}\left(x - \frac{1}{\sqrt{2}}\right) \Rightarrow 3x + 2y - 3\sqrt{2} = 0$$

Normal (multiply through by $6\sqrt{2}$ to clear fractions):

$$y - \frac{3}{2\sqrt{2}} = \frac{2}{3}\left(x - \frac{1}{\sqrt{2}}\right) \Rightarrow 4\sqrt{2}\,x - 6\sqrt{2}\,y + 5 = 0$$

Check of the normal's constant: $6\sqrt{2}\cdot\frac{3}{2\sqrt{2}} = 9$ and $6\sqrt{2}\cdot\frac{2}{3}\cdot\frac{1}{\sqrt{2}} = 4$, so $6\sqrt{2}\,y - 9 = 4\sqrt{2}\,x - 4$, i.e. $4\sqrt{2}x - 6\sqrt{2}y + 5 = 0$. ✓

Example 4 — tangent parallel to a line [PYQ Jul 2024 Q9; Jul 2025 Q10, 2 marks]

Find the points on $y = x^3 - 6x^2 + x + 3$ where the tangents are parallel to $y = x + 5$.
The line $y = x + 5$ has slope 1. Parallel lines have equal slopes, so set $\frac{dy}{dx} = 1$:

$$3x^2 - 12x + 1 = 1 \Rightarrow 3x(x-4) = 0 \Rightarrow x = 0 \text{ or } x = 4$$

$x = 0$: $y = 3$. $\quad x = 4$: $y = 64 - 96 + 4 + 3 = -25$.
Required points: $(0, 3)$ and $(4, -25)$.

The Jul 2025 version asked you to *show* that the tangent at $(0,3)$ is parallel to $y = x + 5$. Same work, reversed: compute $\left(\frac{dy}{dx}\right)_{(0,3)} = 3(0) - 0 + 1 = 1$, which equals the slope of the line, hence parallel.

2.5 More exercise patterns (Ex. 2.1, Q1–Q6, Misc. 2)

Tangent perpendicular to a line (Ex. 2.1 Q2)

Find the point on $y = \sqrt{x-3}$ where the tangent is perpendicular to $6x + 3y - 5 = 0$.
Slope of the line $= -\frac{6}{3} = -2$. Perpendicular slope $= \frac{1}{2}$.

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x-3}} = \frac{1}{2} \Rightarrow \sqrt{x-3} = 1 \Rightarrow x = 4,\ y = 1$$

Point: $(4, 1)$.

Tangents parallel to the X-axis (Ex. 2.1 Q4)

Find the tangents to $x^2 + y^2 - 2x - 4y + 1 = 0$ that are parallel to the X-axis.
$2x + 2y\dfrac{dy}{dx} - 2 - 4\dfrac{dy}{dx} = 0 \Rightarrow \dfrac{dy}{dx} = \dfrac{1-x}{y-2}$.
Parallel to X-axis $\Rightarrow \dfrac{dy}{dx} = 0 \Rightarrow x = 1$. Substitute in the curve: $1 + y^2 - 2 - 4y + 1 = 0 \Rightarrow y(y-4) = 0 \Rightarrow y = 0$ or 4.
Points $(1,0)$, $(1,4)$; tangents $y = 0$ and $y = 4$.

The curve is a circle with centre $(1,2)$ and radius 2, so the horizontal tangents touch it at the bottom $(y = 0)$ and top $(y = 4)$. Geometry confirms the calculus.

Normals parallel to a line (Ex. 2.1 Q5)

Find the equations of the normals to $3x^2 - y^2 = 8$ parallel to $x + 3y = 4$.
$6x - 2y\dfrac{dy}{dx} = 0 \Rightarrow \dfrac{dy}{dx} = \dfrac{3x}{y}$, so slope of normal $= -\dfrac{y}{3x}$.
Slope of $x + 3y = 4$ is $-\frac{1}{3}$. Set $-\dfrac{y}{3x} = -\dfrac{1}{3} \Rightarrow y = x$.
Substitute: $3x^2 - x^2 = 8 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$. Points $(2,2)$ and $(-2,-2)$.
Normal at $(2,2)$: $y - 2 = -\frac{1}{3}(x-2) \Rightarrow x + 3y - 8 = 0$.
Normal at $(-2,-2)$: $y + 2 = -\frac{1}{3}(x+2) \Rightarrow x + 3y + 8 = 0$.

Line touching a curve: find constants (Ex. 2.1 Q6)

If $y = 4x - 5$ touches $y^2 = ax^3 + b$ at $(2,3)$, find a and b.
"Touches at $(2,3)$" gives **two** conditions: the point is on the curve, and the curve's slope there equals the line's slope 4.
Slope: $2y\dfrac{dy}{dx} = 3ax^2 \Rightarrow \left(\dfrac{dy}{dx}\right)_{(2,3)} = \dfrac{3a(4)}{2(3)} = 2a = 4 \Rightarrow a = 2$.
Point on curve: $9 = 2(8) + b \Rightarrow b = -7$.

Tangent and normal for an implicit quartic (Misc. II Q3)

Find the tangent and normal to $y^4 - 4x^4 - 6xy = 0$ at $M(1,2)$.
$4y^3\dfrac{dy}{dx} - 16x^3 - 6\left(x\dfrac{dy}{dx} + y\right) = 0 \Rightarrow \dfrac{dy}{dx} = \dfrac{16x^3 + 6y}{4y^3 - 6x}$.
At $(1,2)$: $m = \dfrac{16+12}{32-6} = \dfrac{28}{26} = \dfrac{14}{13}$.
Tangent: $y - 2 = \frac{14}{13}(x-1) \Rightarrow 14x - 13y + 12 = 0$. Normal: $y - 2 = -\frac{13}{14}(x-1) \Rightarrow 13x + 14y - 41 = 0$.

Triangle formed by a tangent and the axes (Misc. II Q2)

Find the area of the triangle formed by the tangent to $y = 3 - x^2$ at $(1,2)$ and the coordinate axes.
$\dfrac{dy}{dx} = -2x \Rightarrow m = -2$. Tangent: $y - 2 = -2(x-1) \Rightarrow 2x + y = 4$.
Intercepts: X-axis at $(2,0)$, Y-axis at $(0,4)$. Area $= \frac{1}{2} \times 2 \times 4 = 4$ sq. units.

Orthogonal curves (Misc. II Q1) — a proof-type question

Claim. If $ax^2 + by^2 = 1$ and $a'x^2 + b'y^2 = 1$ cut orthogonally, then $\frac{1}{a} - \frac{1}{b} = \frac{1}{a'} - \frac{1}{b'}$.

Proof. Let (x, y) be a point of intersection. Differentiating each curve:

$$m_1 = -\frac{ax}{by}, \qquad m_2 = -\frac{a'x}{b'y}$$

Orthogonal means the tangents are perpendicular: $m_1 m_2 = -1$, so

$$\frac{aa'x^2}{bb'y^2} = -1 \;\Rightarrow\; \frac{x^2}{y^2} = -\frac{bb'}{aa'} \qquad \text{(i)}$$

Subtracting the two curve equations: $(a - a')x^2 + (b - b')y^2 = 0$, so

$$\frac{x^2}{y^2} = \frac{b' - b}{a - a'} \qquad \text{(ii)}$$

Equating (i) and (ii): $aa'(b' - b) = -bb'(a - a')$. Divide both sides by $aa'bb'$:

$$\frac{b' - b}{bb'} = -\frac{a - a'}{aa'} \;\Rightarrow\; \frac{1}{b} - \frac{1}{b'} = \frac{1}{a} - \frac{1}{a'} \;\Rightarrow\; \frac{1}{a} - \frac{1}{b} = \frac{1}{a'} - \frac{1}{b'}. \qquad \blacksquare$$

3 Derivative as a Rate Measure (Textbook 2.1.3)

3.1 Intuition: average vs instantaneous rate

If $y = f(x)$ and x changes from x_1 to x_2, the **average rate of change** is

$$\frac{\delta y}{\delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1},$$

the slope of the chord (secant) joining the two points. Let $x_2 \to x_1$ and the chord turns into the tangent; the limit is the **instantaneous rate of change** $f'(x_1)$.

Your car's trip computer shows *average* speed (total distance ÷ total time). The speedometer shows *instantaneous* speed — that is the derivative.

3.2 Related rates: the method

Most rate problems involve two or more quantities that all change with time t, linked by a geometric formula. The chain rule then links their rates.

Five-step method for every related-rates question

1. **Name** every changing quantity with a letter (they are all functions of t). Draw a figure if there is geometry.
2. **Write what is given** as a rate ($\frac{dr}{dt} = 5$) and **what is asked** ($\frac{dA}{dt} = ?$) and the instant ("when $r = 8$").
3. **Find a formula** linking the quantities (area, volume, Pythagoras, similar triangles). Use fixed constants to eliminate unwanted variables *before* differentiating.

4. **Differentiate w.r.t.** t using the chain rule: $\frac{d}{dt}(r^2) = 2r\frac{dr}{dt}$.
5. **Substitute the instant's values only now**, solve, and write the answer with **units** and the words "increasing" or "decreasing".

The classic trap

Never substitute a value that *changes* (like $r = 8$) before differentiating — it turns a variable into a constant and its derivative into 0. Only genuine constants (like a fixed base radius of 3 m, or a fixed $\tan 30°$) may be substituted before differentiating.

Units: if area is in cm² and time in sec, $\frac{dA}{dt}$ is in cm²/sec. The textbook itself sometimes writes "cc/sec" for a radius rate (Ex. 2, p. 69); that is a misprint — a rate of radius is in cm/sec.

3.3 Useful formulas you will need

Shape	Formula(s)
Circle, radius r	$A = \pi r^2$, circumference $C = 2\pi r$
Sphere, radius r	$V = \frac{4}{3}\pi r^3$, $S = 4\pi r^2$
Cylinder, radius r, height h	$V = \pi r^2 h$, curved surface $2\pi rh$, total $2\pi rh + 2\pi r^2$
Cone, radius r, height h, slant l	$V = \frac{1}{3}\pi r^2 h$, $l^2 = r^2 + h^2$, $\tan(\text{semi-vertical angle}) = \frac{r}{h}$
Cube, edge x	$V = x^3$, $S = 6x^2$
Equilateral triangle, side a	$A = \frac{\sqrt{3}}{4}a^2$
Triangle with sides b, c and included angle A	$\Delta = \frac{1}{2}bc\sin A$

3.4 Worked examples

Circular wave [PYQ Feb 2026 Q12, 2 marks — same as textbook Ex. 1 with changed numbers]

A stone dropped into a quiet lake makes circular waves. The radius increases at 3 cm/sec. How fast is the enclosed area increasing when the radius is 8 cm?
Let R = radius, A = area at time t. $A = \pi R^2$.
Differentiate w.r.t. t: $\frac{dA}{dt} = 2\pi R\frac{dR}{dt}$.
Given $\frac{dR}{dt} = 3$; when $R = 8$: $\frac{dA}{dt} = 2\pi(8)(3) = 48\pi$.
The area is increasing at 48π cm²/sec.

(The textbook version uses 5 cm/sec, giving 80π cm²/sec.)

Sphere: volume rate given (textbook Ex. 2)

The volume of a spherical ball increases at 4π cc/sec. Find the rates at which the radius and surface area change when the volume is 288π cc.

$V = \frac{4}{3}\pi R^3 \Rightarrow \dfrac{dV}{dt} = 4\pi R^2 \dfrac{dR}{dt}$. With $\dfrac{dV}{dt} = 4\pi$: $\dfrac{dR}{dt} = \dfrac{1}{R^2}$.
When $V = 288\pi$: $\frac{4}{3}\pi R^3 = 288\pi \Rightarrow R^3 = 216 \Rightarrow R = 6$. So $\dfrac{dR}{dt} = \dfrac{1}{36}$ cm/sec.
$S = 4\pi R^2 \Rightarrow \dfrac{dS}{dt} = 8\pi R \dfrac{dR}{dt} = 8\pi(6)\left(\frac{1}{36}\right) = \dfrac{4\pi}{3}$ cm²/sec.

Water poured into a cylinder [PYQ Jul 2025 Q21, 3 marks]

Water is poured at 27 m³/sec into a cylindrical vessel of base radius 3 m. Find the rate at which the water level rises.
Let H = height of water, V = volume. The radius is *fixed* at 3, so substitute it now:

$$V = \pi(3)^2 H = 9\pi H \;\Rightarrow\; \frac{dV}{dt} = 9\pi \frac{dH}{dt}$$

$$27 = 9\pi \frac{dH}{dt} \;\Rightarrow\; \frac{dH}{dt} = \frac{3}{\pi} \text{ m/sec}$$

The water level rises at $\dfrac{3}{\pi}$ m/sec (textbook Ex. 3 uses 36 m³/sec, giving $\dfrac{4}{\pi}$ m/sec).

Lamp-post and shadow (textbook Ex. 4)

A man 180 cm tall walks away from a 4.5 m lamp post at 1.2 m/sec. Find the rate at which (i) his shadow lengthens, (ii) the tip of the shadow moves.

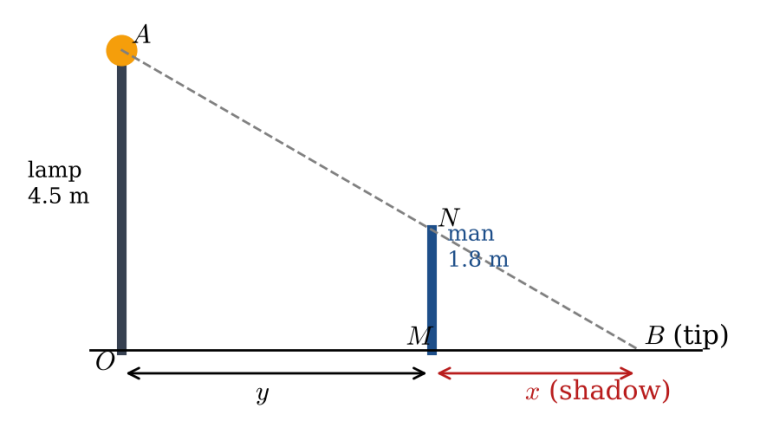


Figure 2: Lamp post OA, man MN, shadow $MB = x$, distance $OM = y$

Triangles BMN and BOA are similar (both right-angled, common angle at B):

$$\frac{MN}{OA} = \frac{BM}{BO} \;\Rightarrow\; \frac{1.8}{4.5} = \frac{x}{x+y} \;\Rightarrow\; 1.8x + 1.8y = 4.5x \;\Rightarrow\; x = \frac{2y}{3}$$

Differentiate: $\dfrac{dx}{dt} = \dfrac{2}{3}\dfrac{dy}{dt} = \dfrac{2}{3}(1.2) = 0.8$ m/sec.
Tip of the shadow is at distance $x + y$ from the post: $\dfrac{d}{dt}(x+y) = 0.8 + 1.2 = 2$ m/sec.
Shadow lengthens at 0.8 m/sec; its tip moves at 2 m/sec.

Notice the shortcut: the rate is constant — it does not depend on where the man is. That is why no "when he is ... m away" is given.

Man walking *towards* the lamp (Ex. 2.1 Q14)

Man 1.5 m, lamp 4.5 m, walking towards it at $\frac{3}{4}$ m/sec.
$\frac{1.5}{4.5} = \frac{x}{x+y} \Rightarrow 3x = x+y \Rightarrow x = \frac{y}{2}$. Walking towards the post means y decreases: $\frac{dy}{dt} = -\frac{3}{4}$.
$\frac{dx}{dt} = \frac{1}{2}\left(-\frac{3}{4}\right) = -\frac{3}{8}$: the shadow is shortening at $\frac{3}{8}$ m/sec.
$\frac{d}{dt}(x+y) = -\frac{3}{8} - \frac{3}{4} = -\frac{9}{8}$: the tip moves towards the post at $\frac{9}{8}$ m/sec.

Sliding ladder (Ex. 2.1 Q15)

A 10 m ladder leans on a wall. The bottom is pulled away at 1.2 m/sec. How fast does the top slide down when the bottom is 6 m from the wall?
Let x = distance of foot from wall, y = height of top. Pythagoras: $x^2 + y^2 = 100$.
Differentiate: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}$.
When $x = 6$: $y = \sqrt{100-36} = 8$. $\frac{dy}{dt} = -\frac{6}{8}(1.2) = -0.9$.
The top slides down at 0.9 m/sec. (Negative sign = height decreasing.)

Water in an inverted cone (Ex. 2.1 Q16)

Water is poured into an inverted cone of semi-vertical angle 30°; depth increases at 1 cm/sec. Find the rate of increase of volume when the depth is 2 cm.
For water depth h and surface radius r: $\tan 30^\circ = \frac{r}{h} \Rightarrow r = \frac{h}{\sqrt{3}}$ (a fixed ratio, so substitute *before* differentiating).

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\frac{h^2}{3}h = \frac{\pi h^3}{9} \;\Rightarrow\; \frac{dV}{dt} = \frac{\pi h^2}{3}\frac{dh}{dt} = \frac{\pi(4)}{3}(1) = \frac{4\pi}{3}\ \text{cm}^3/\text{sec}$$

Tank being emptied (Misc. II Q4)

An inverted conical tank (height 8 ft, radius 4 ft) is emptied at 2 ft³/sec. Find the rate of change of water level when depth is 6 ft.
Similar triangles: $\frac{r}{h} = \frac{4}{8} \Rightarrow r = \frac{h}{2}$. $V = \frac{1}{3}\pi\frac{h^2}{4}h = \frac{\pi h^3}{12}$.
$\frac{dV}{dt} = \frac{\pi h^2}{4}\frac{dh}{dt}$. Being emptied: $\frac{dV}{dt} = -2$. At $h = 6$: $-2 = 9\pi\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = -\frac{2}{9\pi}$ ft/sec.
The level falls at $\frac{2}{9\pi}$ ft/sec.

Point on a curve where y changes 8 times as fast as x (Ex. 2.1 Q7)

A particle moves along $6y = x^3 + 2$. Find the points where y changes 8 times as fast as x.
$6\frac{dy}{dt} = 3x^2\frac{dx}{dt}$. Condition: $\frac{dy}{dt} = 8\frac{dx}{dt}$, so $48\frac{dx}{dt} = 3x^2\frac{dx}{dt} \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$.
$x = 4$: $y = \frac{66}{6} = 11$. $\quad x = -4$: $y = \frac{-62}{6} = -\frac{31}{3}$. Points $(4, 11)$ and $\left(-4, -\frac{31}{3}\right)$.

Ellipse: y decreasing at the same rate x increases (Misc. II Q5)

Find points on $9x^2 + 16y^2 = 400$ where y decreases and x increases at the same rate.
Condition: $\frac{dy}{dt} = -\frac{dx}{dt}$, i.e. $\frac{dy}{dx} = -1$. From the curve: $18x + 32y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{9x}{16y}$.
$-\frac{9x}{16y} = -1 \Rightarrow x = \frac{16y}{9}$. Substitute: $9 \cdot \frac{256y^2}{81} + 16y^2 = 400 \Rightarrow \frac{400y^2}{9} = 400 \Rightarrow y = \pm 3$.
Points $\left(\frac{16}{3}, 3\right)$ and $\left(-\frac{16}{3}, -3\right)$.

Quick answers to the remaining Ex. 2.1 rate questions (verify each yourself):

Q	Data	Answer
8	soap bubble, $\frac{dr}{dt} = 0.02$, $r = 5$	$\frac{dS}{dt} = 8\pi r\frac{dr}{dt} = 0.8\pi$ cm²/sec
9	balloon, $\frac{dS}{dt} = 2$, $r = 6$	$\frac{dr}{dt} = \frac{1}{24\pi}$, $\frac{dV}{dt} = 6$ cm³/sec
10	equilateral $\triangle$, $\frac{da}{dt} = \sqrt{2}$, $a = 3$	$\frac{dA}{dt} = \frac{\sqrt{3}}{2}a\frac{da}{dt} = \frac{3\sqrt{6}}{2}$ cm²/sec
11	sphere, $\frac{dV}{dt} = 20$, $r = 5$	$\frac{dS}{dt} = 8$ cm²/sec
12	cube, $\frac{dx}{dt} = -0.6$, $x = 2$	$\frac{dV}{dt} = -7.2$: decreasing at 7.2 cm³/sec
13	man 2 m, lamp 6 m, 6 km/hr	shadow lengthens at 3 km/hr

4 Velocity, Acceleration and Jerk (Textbook 2.1.4)

If $s = f(t)$ is the displacement of a particle moving on a straight line at time t:

Motion in a straight line

$$v = \frac{ds}{dt}, \qquad a = \frac{dv}{dt} = \frac{d^2s}{dt^2}, \qquad j = \frac{da}{dt} = \frac{d^3s}{dt^3}$$

Speed $= |v|$. A particle is momentarily at rest when $v = 0$. Constant a means **uniformly accelerated motion**.

Jerk is the rate of change of acceleration — the "jolt" you feel when a bus driver brakes suddenly.

Particle with cubic displacement [PYQ Jul 2024 Q11 (2 marks); Feb 2025 Q22 (3 marks); MCQ Jul 2025]

$s = 2t^3 - 5t^2 + 4t - 3$. Find the time when acceleration is 14 ft/sec², and the velocity and displacement then.

$$v = \frac{ds}{dt} = 6t^2 - 10t + 4, \qquad a = \frac{dv}{dt} = 12t - 10$$

$a = 14 \Rightarrow 12t = 24 \Rightarrow t = 2$ sec.
$v(2) = 24 - 20 + 4 = 8$ ft/sec. $\quad s(2) = 16 - 20 + 8 - 3 = 1$ ft.

The Jul 2025 MCQ asked for the acceleration at $t = 1$: $a(1) = 12 - 10 = 2$, option (a).

Car (textbook Ex. 1)

$s = 4t^2 + 3t$ m. Then $v = 8t + 3$, $a = 8$. At $t = 20$: $v = 163$ m/sec, $a = 8$ m/sec² (independent of t: uniformly accelerated).

VSA [PYQ Jun 2026 Q2(iii), 1 mark]

$S = 3t^2 + 4t + 1$. Velocity $v = \dfrac{dS}{dt} = 6t + 4$.

Average velocity = instantaneous velocity (Misc. II Q7)

$s(t) = 2t^2 + 3t - 4$, $0 \le t \le 4$. Average velocity $= \dfrac{s(4) - s(0)}{4 - 0} = \dfrac{40 - (-4)}{4} = 11$. Instantaneous $v = 4t + 3 = 11 \Rightarrow t = 2$. So $c = 2$.
This is LMVT in disguise: some instant has velocity equal to the average velocity.

5 Approximations (Textbook 2.2.1)

5.1 Intuition

Near the point $x = a$, a smooth curve is almost indistinguishable from its tangent line. So to estimate f at a nearby point $a + h$, walk along the **tangent** instead of the curve: start at height $f(a)$ and rise by (slope $\times$ run) $= f'(a) \cdot h$.

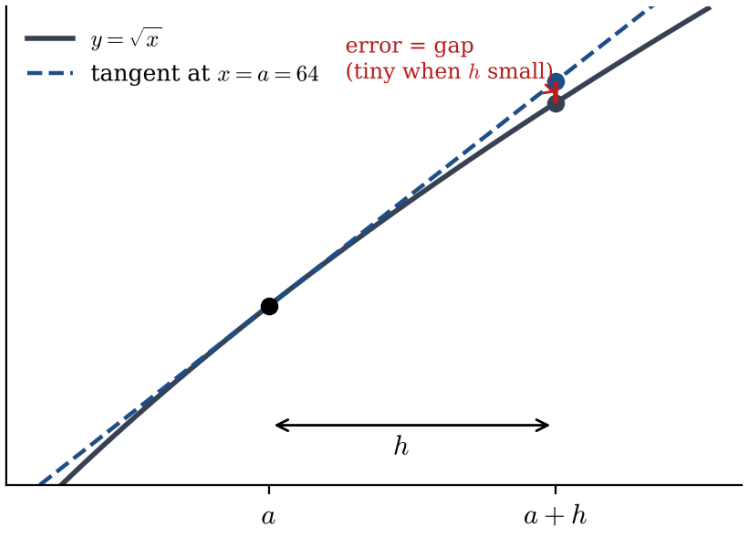


Figure 3: The tangent line at $x = a$ hugs the curve; the gap (error) is tiny when h is small

5.2 Derivation

By definition, $f'(a) = \lim\limits_{h \to 0} \dfrac{f(a+h) - f(a)}{h}$. For a sufficiently small h the quotient is close to its limit:

$$f'(a) \approx \frac{f(a+h) - f(a)}{h} \;\Rightarrow\; hf'(a) \approx f(a+h) - f(a)$$

Approximation formula

$$f(a+h) \approx f(a) + h\,f'(a)$$

Choose a so that $f(a)$ and $f'(a)$ are **easy to compute exactly**, and $h = (\text{given value}) - a$ is **small**. h may be negative.

5.3 How to choose a and h

Asked to approximate	Take $f(x)$	a	h
$\sqrt{64.1}$	$\sqrt{x}$	64	0.1
$(3.98)^3$	x^3	4	-0.02
$\sqrt[3]{28}$	$x^{1/3}$	27	1
$\sin(30°30')$	$\sin x$	$\frac{\pi}{6}$	$30' = \frac{1}{2}° = 0.00875^c$
$\tan^{-1}(0.99)$	$\tan^{-1} x$	1	-0.01
$e^{1.005}$	e^x	1	0.005
$\log_{10} 998$	$\log_{10} x$	1000	-2

Two errors examiners see every year

1. **Degrees in trigonometry.** h must be in **radians**. Use the given $1° = 0.0175^c$; so $30' = \frac{1}{2}° = 0.00875^c$ and $40' = \frac{2}{3}° = 0.011667^c$. Using $h = 0.5$ is wrong by a factor of 57.
2. **Writing "=" instead of "≈".** The answer is approximate. The textbook uses the symbol ≒; "≈" is equally acceptable.

5.4 Worked examples (textbook Ex. 1–7, p. 73–75)

$\sqrt{64.1}$

$f(x) = \sqrt{x}$, $f'(x) = \dfrac{1}{2\sqrt{x}}$; $a = 64$, $h = 0.1$. $f(64) = 8$, $f'(64) = \dfrac{1}{16} = 0.0625$.

$$\sqrt{64.1} \approx 8 + (0.1)(0.0625) = 8.00625$$

$(3.98)^3$

$f(x) = x^3$, $f'(x) = 3x^2$; $a = 4$, $h = -0.02$. $f(4) = 64$, $f'(4) = 48$.

$$(3.98)^3 \approx 64 + (-0.02)(48) = 64 - 0.96 = 63.04$$

(Actual value 63.044 . . . — the error is in the third decimal.)

$\sin(30°30')$, given $1° = 0.0175^c$, $\cos 30° = 0.866$

$f(x) = \sin x$, $f'(x) = \cos x$; $a = \dfrac{\pi}{6}$, $h = \dfrac{1°}{2} = \dfrac{0.0175}{2} = 0.00875^c$.

$$\sin(30°30') \approx \sin 30° + (0.00875)\cos 30° = 0.5 + (0.00875)(0.866) = 0.5 + 0.0075775 = 0.5075775$$

$\tan^{-1}(1.002)$, given $\pi = 3.1416$ [PYQ Mar 2024 Q22, 3 marks]

$f(x) = \tan^{-1} x$, $f'(x) = \dfrac{1}{1+x^2}$; $a = 1$, $h = 0.002$.

$f(1) = \tan^{-1} 1 = \dfrac{\pi}{4} = \dfrac{3.1416}{4} = 0.7854$; $f'(1) = \dfrac{1}{1+1} = 0.5$.

$$\tan^{-1}(1.002) \approx 0.7854 + (0.002)(0.5) = 0.7854 + 0.001 = 0.7864$$

The textbook does $\tan^{-1}(0.99)$ with $h = -0.01$: $0.7854 - 0.005 = 0.7804$.

$\tan(44°30')$ [PYQ MCQ Jul 2024 = Misc. I Q10]

$f(x) = \tan x$, $f'(x) = \sec^2 x$; $a = 45° = \dfrac{\pi}{4}$, $h = -30' = -0.00875^c$. $f(a) = 1$, $f'(a) = \sec^2 \dfrac{\pi}{4} = 2$.

$$\tan(44°30') \approx 1 + (-0.00875)(2) = 1 - 0.0175 = 0.9825 \quad \text{— option (d)}$$

$e^{1.005}$, given $e = 2.7183$

$f(x) = e^x = f'(x)$; $a = 1$, $h = 0.005$. $e^{1.005} \approx 2.7183 + (0.005)(2.7183) = 2.7183 + 0.0135915 = 2.7318915 \approx 2.73189$.

$\log_{10} 998$, given $\log_{10} e = 0.4343$

$f(x) = \log_{10} x = \dfrac{\log x}{\log 10} = (\log_{10} e)\log x$, so $f'(x) = \dfrac{\log_{10} e}{x} = \dfrac{0.4343}{x}$.

$a = 1000$, $h = -2$: $f(1000) = 3$, $f'(1000) = 0.0004343$.

$$\log_{10} 998 \approx 3 - 2(0.0004343) = 3 - 0.0008686 = 2.9991314$$

Polynomial at $x = 1.98$ [PYQ Jul 2024 Q22, 3 marks]

$f(x) = x^3 + 5x^2 - 2x + 3$, $f'(x) = 3x^2 + 10x - 2$; $a = 2$, $h = -0.02$.

$f(2) = 8 + 20 - 4 + 3 = 27$; $f'(2) = 12 + 20 - 2 = 30$.

$$f(1.98) \approx 27 + (-0.02)(30) = 27 - 0.6 = 26.4$$

MCQ [PYQ Feb 2026 Q1(vi)]: $f(x) = x^3 - 3x + 5$ at $x = 1.99$

$a = 2$, $h = -0.01$; $f(2) = 7$, $f'(x) = 3x^2 - 3 \Rightarrow f'(2) = 9$. $f(1.99) \approx 7 - 0.09 = 6.91$ — option (b).

$\cos^{-1}(0.51)$, given $\pi = 3.1416$, $\frac{2}{\sqrt{3}} = 1.1547$ (Misc. II Q9)

$f(x) = \cos^{-1} x$, $f'(x) = -\dfrac{1}{\sqrt{1-x^2}}$; $a = 0.5$, $h = 0.01$.

$f(0.5) = \dfrac{\pi}{3} = 1.0472$; $f'(0.5) = -\dfrac{1}{\sqrt{3/4}} = -\dfrac{2}{\sqrt{3}} = -1.1547$.

$$\cos^{-1}(0.51) \approx 1.0472 - (0.01)(1.1547) = 1.0472 - 0.011547 = 1.035653$$

Answers to Ex. 2.2 for self-checking:

Value	Answer	Value	Answer
$\sqrt{8.95}$	2.99167	$\sqrt[3]{28}$	3.0370
$\sqrt[5]{31.98}$	1.99975	$(3.97)^4$	248.32
$(4.01)^3$	64.48	$\sin 61^\circ$	0.87475
$\sin(29^\circ 30')$	0.4924225	$\cos(60^\circ 30')$	0.4924225
$\tan(45^\circ 40')$	1.023333	$\tan^{-1}(0.999)$	0.7849
$\cot^{-1}(0.999)$	0.7859	$\tan^{-1}(1.001)$	0.7859
$e^{0.995}$	2.7047	$e^{2.1}$	8.1279
$3^{2.01}$	9.098874	$\log_e 101$	4.6152
$\log_e 9.01$	2.198311	$\log_{10} 1016$	3.0069488
$x^3 + 5x^2 - 7x + 10$ at 1.12	9.72	$\sqrt{x^2 + 3x}$ at 1.02	2.025

(For $3^{2.01}$: $f(x) = 3^x$, $f'(x) = 3^x \log 3$, $a = 2$, $h = 0.01$: $9 + 0.01(9)(1.0986) = 9.098874$.)

6 Rolle's Theorem and Lagrange's Mean Value Theorem (Textbook 2.3)

6.1 Rolle's Theorem

Rolle's Theorem

If a real-valued function f is

1. **continuous** on the closed interval $[a, b]$,
2. **differentiable** on the open interval (a, b), and
3. $f(a) = f(b)$,

then there exists **at least one** $c \in (a, b)$ such that $f'(c) = 0$.

Intuition. You drive from home and come back home. Your position started and ended at the same value. Somewhere in between you must have turned around — at that instant your velocity was zero. Rolle's theorem is that statement for graphs.

Geometrical meaning. If the graph is unbroken (continuity), smooth with a tangent everywhere inside (differentiability), and the end-points are at the same height, then somewhere between them the tangent is **parallel to the X-axis**.

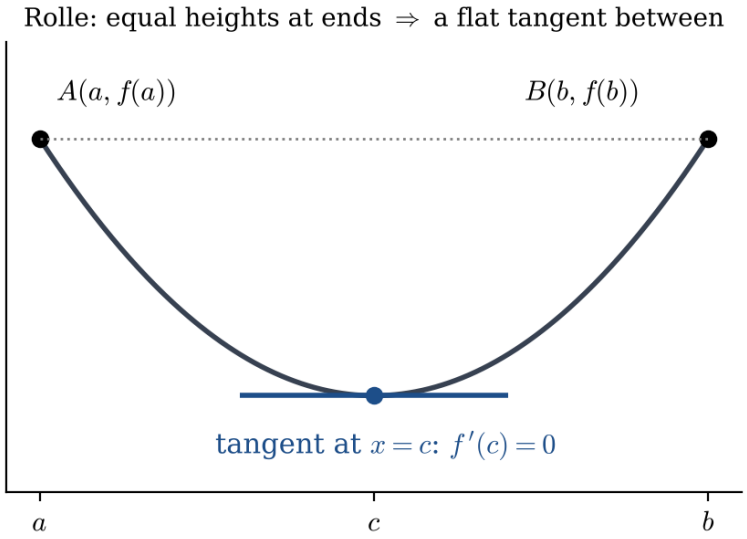


Figure 4: Rolle's theorem for $f(x) = x^2 - 4x + 10$ on $[0, 4]$: $c = 2$

Why each condition matters

A theorem's conditions are not decoration. Remove any one and the conclusion can fail:

Condition dropped	Counter-example	What goes wrong
Continuity on $[a, b]$	$f(x) = x$ on $[0, 1)$, $f(1) = 0$	jump at $x = 1$; $f'(x) = 1 \neq 0$ everywhere
Differentiability on (a, b)	$f(x) = \lvert x \rvert$ on $[-1, 1]$	sharp corner at 0; $f' = \pm 1$, never 0
$f(a) = f(b)$	$f(x) = x$ on $[0, 1]$	always rising; $f' = 1$

Also: the theorem guarantees *at least one* c. There may be more (see Example below with two values of c).

Which functions are continuous and differentiable?

You must *state* these conditions in answers. Standard facts you may quote:

- **Polynomials** are continuous and differentiable everywhere.
- $\sin x$, $\cos x$, e^x are continuous and differentiable everywhere; $\log x$ for $x > 0$; a quotient wherever its denominator $\neq 0$; $\sqrt{g(x)}$ is differentiable where $g(x) > 0$.
- Sums, products and compositions of such functions inherit these properties on the common domain.
- **Danger signs:** $|x|$, fractional powers like $x^{2/3}$ (derivative blows up at 0), piecewise definitions (check the joining point), denominators vanishing inside $[a, b]$.

Method for "verify Rolle's theorem"

Answer template (3 marks)

1. State: f is continuous on $[a, b]$ and differentiable on (a, b) — with the reason (polynomial / composition of continuous functions ...).
2. Compute $f(a)$ and $f(b)$; show they are equal.
3. Conclude: "all conditions of Rolle's theorem are satisfied".
4. Find $f'(x)$, solve $f'(c) = 0$.
5. Show $c \in (a, b)$. Write "Hence Rolle's theorem is verified."

Verify Rolle's theorem: $f(x) = x^2 - 5x + 9$ on $[1, 4]$ [PYQ Jun 2026 Q23, 3 marks]

$f(x)$ is a polynomial, hence continuous on $[1, 4]$ and differentiable on $(1, 4)$.
$f(1) = 1 - 5 + 9 = 5$; $f(4) = 16 - 20 + 9 = 5$. So $f(1) = f(4)$.
All conditions of Rolle's theorem are satisfied. $f'(x) = 2x - 5$.
$f'(c) = 0 \Rightarrow 2c - 5 = 0 \Rightarrow c = \dfrac{5}{2} = 2.5 \in (1, 4)$. Hence Rolle's theorem is verified.

Check the conditions: $f(x) = x^2 - 4x + 3$ on $[1, 3]$ [PYQ Jul 2025 Q11, 2 marks]

Polynomial $\Rightarrow$ continuous on $[1, 3]$, differentiable on $(1, 3)$. $f(1) = 1 - 4 + 3 = 0$, $f(3) = 9 - 12 + 3 = 0$, so $f(1) = f(3)$.
Hence the conditions of Rolle's theorem are satisfied. (Here $c = 2$, from $2c - 4 = 0$.)

Conditions *not* satisfied (textbook Ex. 1(ii))

$f(x) = x^2 - 2x + 3$ on $[1, 4]$: polynomial, so conditions 1–2 hold, but $f(1) = 2$ and $f(4) = 11$. Since $f(1) \neq f(4)$, Rolle's theorem is not applicable.

Two values of c (textbook Ex. 1(i))

$f(x) = 2x^3 - 5x^2 + 3x + 2$ on $\left[0, \frac{3}{2}\right]$. Polynomial; $f(0) = 2$, $f\left(\frac{3}{2}\right) = \frac{27}{4} - \frac{45}{4} + \frac{9}{2} + 2 = 2$. Conditions hold.

$f'(c) = 6c^2 - 10c + 3 = 0 \Rightarrow c = \dfrac{10 \pm \sqrt{100-72}}{12} = \dfrac{5 \pm \sqrt{7}}{6}$, i.e. $c \approx 0.392$ or $c \approx 1.274$. **Both** lie in $\left(0, \frac{3}{2}\right)$.

Exponential–trig: $f(x) = e^x(\sin x - \cos x)$ on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$ (textbook Ex. 4)

e^x, $\sin x$, $\cos x$ are continuous and differentiable everywhere, so f is too.
$f\left(\frac{\pi}{4}\right) = e^{\pi/4}\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) = 0$; $f\left(\frac{5\pi}{4}\right) = e^{5\pi/4}\left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) = 0$.
$f'(x) = e^x(\cos x + \sin x) + e^x(\sin x - \cos x) = 2e^x \sin x$. Since $e^c \neq 0$: $\sin c = 0 \Rightarrow c = n\pi$. The only such value in $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ is $c = \pi$. Verified.

Piecewise — fails at the joint (Ex. 2.3 Q1(v))

$f(x) = x^2$ for $0 \le x \le 2$; $f(x) = 6 - x$ for $2 \le x \le 6$.
Continuity at 2: both pieces give 4, so f is continuous. Derivatives at 2: left $= 2x\big|_2 = 4$, right $= -1$. They differ, so f is **not differentiable at** $x = 2$. Rolle's theorem is not applicable (even though $f(0) = 0 = f(6)$).

Fractional power (Ex. 2.3 Q1(vi))

$f(x) = x^{2/3}$ on $[-1, 1]$: $f'(x) = \dfrac{2}{3x^{1/3}}$ does not exist at $x = 0 \in (-1, 1)$. Not differentiable; Rolle's theorem is not applicable.

Find unknown constants from Rolle (Ex. 2.3 Q4; Misc. I Q1)

Rolle's theorem holds for $f(x) = x^3 + px^2 + qx + 5$ on $[1, 3]$ with $c = 2 + \frac{1}{\sqrt{3}}$. Find p, q.
$f(1) = f(3)$: $6 + p + q = 32 + 9p + 3q \Rightarrow 8p + 2q = -26 \Rightarrow q = -13 - 4p$.
$f'(c) = 3c^2 + 2pc + q = 0$. Compute $c^2 = 4 + \frac{4}{\sqrt{3}} + \frac{1}{3}$, so $3c^2 = 13 + 4\sqrt{3}$ and $2pc = 4p + \frac{2p}{\sqrt{3}}$:

$$13 + 4\sqrt{3} + 4p + \frac{2p}{\sqrt{3}} + (-13 - 4p) = 0 \;\Rightarrow\; 4\sqrt{3} + \frac{2p}{\sqrt{3}} = 0 \;\Rightarrow\; p = -6, \quad q = -13 + 24 = 11.$$

The same idea answers Misc. I Q1: $ax^3 + bx^2 + 11x - 6$ with $f'\left(2 + \frac{1}{\sqrt{3}}\right) = 0$ gives $a = 1$, $b = -6$ — option (A).

Existence of a root (Ex. 2.3 Q5)

Rolle's theorem holds for $f(x) = (x - 2)\log x$ on $[1, 2]$. Show $x \log x = 2 - x$ has a root in $(1, 2)$.
$f(1) = (-1)(0) = 0$, $f(2) = 0$. By Rolle, $\exists\, c \in (1, 2)$ with $f'(c) = 0$.

$$f'(x) = \log x + \frac{x-2}{x} = 0 \;\Rightarrow\; x\log x + x - 2 = 0 \;\Rightarrow\; x\log x = 2 - x.$$

So c satisfies the equation. ■

Other Ex. 2.3 answers: Q2 $b = 1$ (even function, $f(-1) = f(1)$); Q3(i) $c = \frac{\pi}{4}, \frac{5\pi}{4}$; Q3(ii) $c = \pi$; Q3(iii) $c = \frac{5}{2}$; Q6 $c = -2$ (from $f'(x) = -\frac{1}{2}e^{-x/2}(x-3)(x+2)$); Misc. II Q6: $c = 0$.

6.2 Lagrange's Mean Value Theorem (LMVT)

Lagrange's Mean Value Theorem

If f is **continuous** on $[a,b]$ and **differentiable** on (a,b), then there exists **at least one** $c \in (a,b)$ such that

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

Intuition. You drive 120 km in 2 hours: average speed 60 km/h. LMVT says that at some instant the speedometer read exactly 60. (This is how average-speed toll cameras justify a fine.)

Geometrical meaning. The right side is the slope of the chord AB joining $(a,f(a))$ and $(b,f(b))$. So somewhere between, the tangent is **parallel to the chord**.

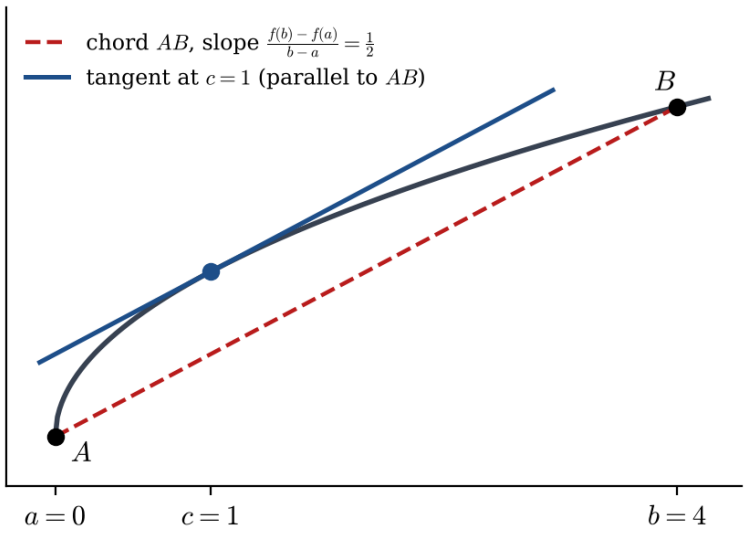


Figure 5: LMVT for $f(x)=\sqrt{x}$ on $[0,4]$: the tangent at $c=1$ is parallel to chord AB

Relation to Rolle. If $f(a)=f(b)$, the chord is horizontal and LMVT becomes Rolle's theorem. Rolle is the special case; LMVT is the tilted version.

Verify LMVT for $f(x)=\log x$ on $[1,e]$ [PYQ Feb 2026 Q22, 3 marks]

$\log x$ is continuous on $[1,e]$ and differentiable on $(1,e)$ (it is so for all $x>0$). So LMVT applies. $f(1)=\log 1=0$, $f(e)=\log e=1$. $f'(x)=\dfrac{1}{x}$.

$$f'(c)=\frac{f(e)-f(1)}{e-1} \Rightarrow \frac{1}{c}=\frac{1-0}{e-1} \Rightarrow c=e-1$$

Since $2<e<3$, $1<e-1<2<e$, so $c=e-1\in(1,e)$. LMVT is verified.

$f(x)=\sqrt{x+4}$ on $[0,5]$ (textbook Ex. 1)

Continuous on $[0,5]$ and differentiable on $(0,5)$ (since $x+4>0$ there). $f(0)=2$, $f(5)=3$, $f'(x)=\dfrac{1}{2\sqrt{x+4}}$.

$$\frac{1}{2\sqrt{c+4}} = \frac{3-2}{5-0} = \frac{1}{5} \Rightarrow \sqrt{c+4} = \frac{5}{2} \Rightarrow c = \frac{25}{4} - 4 = \frac{9}{4} \in (0,5).$$

$f(x) = x + \frac{1}{x}$ on $[1,3]$ — reject the root outside (textbook Ex. 2)

Continuous and differentiable on $[1,3]$ (denominator $x \neq 0$ there). $f(1) = 2$, $f(3) = \frac{10}{3}$, $f'(x) = 1 - \frac{1}{x^2}$.

$$1 - \frac{1}{c^2} = \frac{\frac{10}{3} - 2}{2} = \frac{2}{3} \Rightarrow c^2 = 3 \Rightarrow c = \pm\sqrt{3}.$$

$c = \sqrt{3} \approx 1.732 \in (1,3)$ ✓; $c = -\sqrt{3} \notin (1,3)$, rejected. LMVT verified with $c = \sqrt{3}$.

Shortcut for quadratics (use it to check, not as the written method)

For any quadratic $f(x) = px^2 + qx + r$ on $[a,b]$, the LMVT point is always the **midpoint**: $c = \frac{a+b}{2}$. E.g. Ex. 2.3 Q7(iii): $x^2 - 3x - 1$ on $[-\frac{11}{7}, \frac{13}{7}]$ gives $c = \frac{1}{7}$; Q7(iv): $2x - x^2$ on $[0,1]$ gives $c = \frac{1}{2}$. Proof: $\frac{f(b)-f(a)}{b-a} = p(a+b) + q$ and $f'(c) = 2pc + q$.

Other Ex. 2.3 Q7 answers: (ii) $(x-1)(x-2)(x-3)$ on $[0,4]$: $c = 2 \pm \frac{2}{\sqrt{3}}$ (both in $(0,4)$); (v) $\frac{x-1}{x-3}$ on $[4,5]$: $c = 3 + \sqrt{2}$.

Rolle / LMVT mistakes

- Not *stating* continuity and differentiability with a reason. That is a method mark.
- Finding c but not checking $c \in (a,b)$ — or keeping a root outside the interval.
- Using a closed interval for differentiability. It is **open** (a,b) for differentiability, **closed** $[a,b]$ for continuity.
- Saying "Rolle's theorem is false" when a condition fails. Correct phrasing: "Rolle's theorem is **not applicable**."

7 Increasing and Decreasing Functions (Textbook 2.4.1)

7.1 Definitions

Strictly increasing / decreasing on an interval

f is **strictly increasing** on an interval I if for all $x_1, x_2 \in I$: $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$.
f is **strictly decreasing** on I if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$.

Going left to right, an increasing graph climbs; a decreasing graph falls.

7.2 Link with the derivative

The textbook shows (p. 80–81): if f is increasing, every difference quotient $\dfrac{f(x+h)-f(x)}{h} > 0$, so its limit $f'(x) \geq 0$. Conversely, if $f'(a) > 0$, then in a small neighbourhood $f(a-h) < f(a) < f(a+h)$.

Derivative test (as stated in the textbook's 'Let us Remember')

Condition	Conclusion
$f'(x) > 0$ on an interval	f is **strictly** (monotonically) increasing there
$f'(x) < 0$ on an interval	f is **strictly** decreasing there
$f'(x) \geq 0$	f is increasing
$f'(x) \leq 0$	f is decreasing

A subtle point that the strong student should know

$f'(x) > 0$ is **sufficient**, not necessary, for strictly increasing. $f(x) = x^3$ is strictly increasing on $\mathbb{R}$ although $f'(0) = 0$. Isolated points where $f' = 0$ do not stop a function from being strictly increasing. This is why the textbook accepts $f'(x) = 3(x+2)^2 \geq 0$ as showing $x^3 + 6x^2 + 12x - 5$ is increasing (Ex. 2).

VSA [PYQ Feb 2025 Q2(iv), 1 mark]: *Condition for $f(x)$ to be strictly increasing for all $x \in \mathbb{R}$:* $f'(x) > 0$ for all $x \in \mathbb{R}$.

7.3 Method: finding intervals of increase/decrease

1. Find $f'(x)$ and factorise it fully.
2. Solve $f'(x) = 0$ (and note points where f' is undefined or f is undefined). These **split the real line**.
3. Test the sign of f' in each piece (pick a test value, or use the sign of each factor).
4. $f' > 0$ pieces: increasing. $f' < 0$ pieces: decreasing.

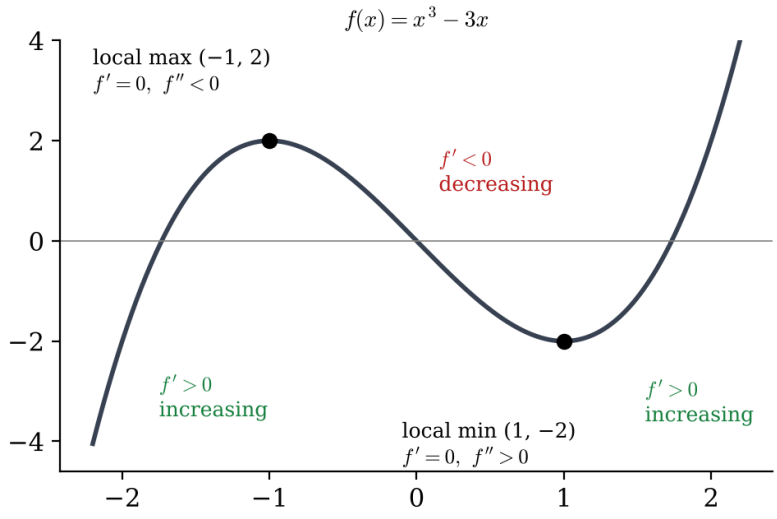


Figure 6: Sign of f' decides the direction; at turning points $f' = 0$

Test for all x [PYQ Feb 2026 Q11, 2 marks]

Test whether $f(x) = x^3 + 6x^2 + 12x - 7$ is increasing or decreasing for all $x \in \mathbb{R}$.
$f'(x) = 3x^2 + 12x + 12 = 3(x^2 + 4x + 4) = 3(x+2)^2$.
$3(x+2)^2 > 0$ for $x \neq -2$ and $= 0$ at $x = -2$, so $f'(x) \geq 0$ for all $x \in \mathbb{R}$. Hence f is an increasing function for all $x \in \mathbb{R}$.

(The textbook Ex. 2 has -5 instead of -7; the constant does not affect f'.)

[PYQ Jun 2026 Q9, 2 marks; Ex. 2.4 Q1(iii)]

Test whether $f(x) = x - \dfrac{1}{x}$, $x \in \mathbb{R}$, $x \neq 0$, is increasing or decreasing.
$f'(x) = 1 + \dfrac{1}{x^2}$. For every $x \neq 0$, $\dfrac{1}{x^2} > 0$, so $f'(x) > 1 > 0$. Hence f is strictly increasing (on each interval of its domain, $(-\infty, 0)$ and $(0, \infty)$).

Find the intervals (textbook Ex. 3)

$f(x) = x^3 + 12x^2 + 36x + 6$. $f'(x) = 3x^2 + 24x + 36 = 3(x+2)(x+6)$.
Critical points $x = -6$, $x = -2$ split $\mathbb{R}$ into three parts:

Interval	sign of $(x+6)$	sign of $(x+2)$	$f'(x)$	f is
$x < -6$	$-$	$-$	$+$	increasing
$-6 < x < -2$	$+$	$-$	$-$	decreasing
$x > -2$	$+$	$+$	$+$	increasing

Increasing on $(-\infty, -6) \cup (-2, \infty)$; decreasing on $(-6, -2)$.

A sign table like this is cleaner than the textbook's case-by-case analysis and earns the same marks.

Rational function (Ex. 2.4 Q6)

$f(x) = \dfrac{x}{x^2+1}$. Quotient rule:

$$f'(x) = \frac{(x^2+1)(1) - x(2x)}{(x^2+1)^2} = \frac{1 - x^2}{(x^2+1)^2}.$$

The denominator is always positive, so the sign is that of $1 - x^2$: increasing for $-1 < x < 1$; decreasing for $x < -1$ or $x > 1$.

Show increasing for all x (Ex. 2.4 Q8)

$f(x) = x - \cos x$: $f'(x) = 1 + \sin x \geq 0$ since $\sin x \geq -1$. Hence increasing for all x.

Show increasing on a domain (Ex. 2.4 Q23)

$y = \log(1+x) - \dfrac{2x}{2+x}$, $x > -1$.

$$\frac{dy}{dx} = \frac{1}{1+x} - \frac{(2+x)2-2x}{(2+x)^2} = \frac{1}{1+x} - \frac{4}{(2+x)^2} = \frac{(2+x)^2-4(1+x)}{(1+x)(2+x)^2} = \frac{x^2}{(1+x)(2+x)^2}.$$

For $x > -1$: numerator ≥ 0, $1+x > 0$, $(2+x)^2 > 0$. So $\frac{dy}{dx} \geq 0$: increasing on its domain.

Other answers (Ex. 2.4 Q2–Q7, Misc. II Q10–11):

Function	Increasing	Decreasing
$2x^3 - 3x^2 - 12x + 6$	$x < -1$ or $x > 2$	$-1 < x < 2$
$3 + 3x - 3x^2 + x^3$	all x ($f' = 3(x-1)^2 \geq 0$)	—
$x^3 - 6x^2 - 36x + 7$	$x < -2$ or $x > 6$	$-2 < x < 6$
$x + \frac{25}{x}$	$x < -5$ or $x > 5$	$-5 < x < 0, 0 < x < 5$
$x^3 - 9x^2 + 24x + 12$	$x < 2$ or $x > 4$	$2 < x < 4$
$x^3 - 12x^2 - 144x + 13$	$x < -4$ or $x > 12$	$-4 < x < 12$
$2x^3 - 15x^2 - 144x - 7$	$x < -3$ or $x > 8$	$-3 < x < 8$
x^x, $x > 0$	$x > \frac{1}{e}$	$0 < x < \frac{1}{e}$
$\frac{x}{\log x}$	$x > e$	$0 < x < 1, 1 < x < e$

8 Maxima and Minima (Textbook 2.4.2 – 2.4.4)

8.1 Definitions

Local maximum and local minimum

f has a **local maximum** at $x = c$ if $f(c) > f(x)$ for all $x \neq c$ in some small interval $(c-\delta, c+\delta)$. Then $f(c)$ is a local maximum value.
f has a **local minimum** at $x = c$ if $f(c) < f(x)$ for all $x \neq c$ near c.

"Local" matters: a local maximum is the top of a hill, not necessarily the highest point of the whole mountain range. A local maximum value can even be smaller than a local minimum value elsewhere.

A point where $f'(c) = 0$ is called a **stationary (turning) point**: the tangent is horizontal. Extreme values occur at such points (for differentiable f), but not every stationary point is an extreme — see points of inflexion.

8.2 First derivative test

First derivative test (h a small positive number)

$f'(c)$	sign of f' just left $(c-h)$	sign just right $(c+h)$	Conclusion
0	+ (rising)	− (falling)	local **maximum**
0	− (falling)	+ (rising)	local **minimum**

0	same sign on both sides	neither — **point of inflexion** (e.g. x^3 at 0)

8.3 Second derivative test

Second derivative test

If $f'(c) = 0$ and

- $f''(c) < 0$: local **maximum** at c (slope is decreasing through zero: $+ \to 0 \to -$).
- $f''(c) > 0$: local **minimum** at c (slope increasing: $- \to 0 \to +$).
- $f''(c) = 0$: the test **fails**; use the first derivative test.

Memory trick: a smiling face $\smile$ has positive curvature and sits at a **min**imum; a frowning face $\frown$ ($f'' < 0$) is at a **max**imum. "f'' positive — cup holds water — minimum."

8.4 Worked examples

Local max/min by first derivative test (textbook Ex. 1)

$f(x) = x^3 - 3x$. $f'(x) = 3(x^2 - 1) = 0 \Rightarrow x = \pm 1$.
At $x = 1$: $f'(1-h) = 3h(h-2) < 0$ and $f'(1+h) = 3h(h+2) > 0$ for small $h > 0$: sign $- \to +$, **local minimum** $f(1) = -2$.
At $x = -1$: $f'(-1-h) = 3h(h+2) > 0$ and $f'(-1+h) = -3h(2-h) < 0$: sign $+ \to -$, **local maximum** $f(-1) = 2$.

Textbook misprints in this example

In Ex. 1 (p. 84) some intermediate lines say "$f'(x) > 0$ … decreasing for $x > 1$" and "$f'(x) < 0$ … increasing for $x < 1$". These are swapped: just left of 1 the function decreases ($f' < 0$) and just right it increases ($f' > 0$). The conclusions (min -2, max 2) are correct. Write the logic correctly in your answer.

Second derivative test (textbook Ex. 1, p. 85)

$f(x) = x^3 - 3x^2 - 24x + 5$. $f'(x) = 3x^2 - 6x - 24 = 3(x+2)(x-4) \Rightarrow x = -2, 4$. $f''(x) = 6x - 6$.
$f''(-2) = -18 < 0$: maximum, $f(-2) = -8 - 12 + 48 + 5 = 33$.
$f''(4) = 18 > 0$: minimum, $f(4) = 64 - 48 - 96 + 5 = -75$.

Ex. 2.4 Q9(ii): $f(x) = 2x^3 - 21x^2 + 36x - 20$

$f'(x) = 6x^2 - 42x + 36 = 6(x-1)(x-6)$; $f''(x) = 12x - 42$.
$f''(1) = -30 < 0$: max $f(1) = 2 - 21 + 36 - 20 = -3$. $f''(6) = 30 > 0$: min $f(6) = 432 - 756 + 216 - 20 = -128$.

Functions with log (Ex. 2.4 Q9(v),(vi))

$f(x) = x\log x$: $f'(x) = \log x + 1 = 0 \Rightarrow x = \frac{1}{e}$; $f''(x) = \frac{1}{x} > 0$: minimum value $f\left(\frac{1}{e}\right) = -\frac{1}{e}$.
$f(x) = \frac{\log x}{x}$: $f'(x) = \frac{1 - \log x}{x^2} = 0 \Rightarrow x = e$; $f''(x) = \frac{2\log x - 3}{x^3}$, $f''(e) = -\frac{1}{e^3} < 0$: maximum value $\frac{1}{e}$.

Trigonometric (Misc. II Q21): $f(x) = \cos^2 x + \sin x$

Write it in terms of $s = \sin x$: $f = 1 - s^2 + s = \frac{5}{4} - \left(s - \frac{1}{2}\right)^2$, with $-1 \le s \le 1$.
Maximum $\frac{5}{4}$ when $\sin x = \frac{1}{2}$; minimum when $\left(s - \frac{1}{2}\right)^2$ is largest, i.e. $s = -1$: $f = 1 - 1 - 1 = -1$.
(By derivatives: $f'(x) = \cos x(1 - 2\sin x) = 0$ gives the same stationary points.)

Other Ex. 2.4 Q9 answers: (i) $y = 5x^3 + 2x^2 - 3x$: max at $x = -\frac{3}{5}$ (value $\frac{36}{25}$), min at $x = \frac{1}{3}$ (value $-\frac{16}{27}$); (iii) $x^3 - 9x^2 + 24x$: max 20 at $x = 2$, min 16 at $x = 4$; (iv) $x^2 + \frac{16}{x^2}$: min 8 at $x = \pm 2$.

9 Optimisation Problems (word problems on maxima and minima)

This is the only part of the chapter that regularly becomes a **4-mark** question. The mathematics is easy; the setting-up is where students lose marks.

Six-step method

1. **Draw and name.** Assign letters to all lengths. Decide what is to be maximised/minimised (the *objective*).
2. **Constraint.** Write the fixed condition (perimeter, area, volume, sum ...) as an equation.
3. **One variable.** Use the constraint to write the objective in terms of a **single** variable.
4. **Differentiate and equate to zero.** Solve; reject values that are meaningless (negative lengths, zero dimensions).
5. **Second derivative test** to confirm maximum or minimum.
6. **Answer the question asked** — dimensions, or the maximum value, with units.

Divide 20 into two parts with minimum sum of squares [PYQ Feb 2025 Q9, 2 marks]

Let the parts be x and $20 - x$. $S = x^2 + (20 - x)^2$.
$\frac{dS}{dx} = 2x - 2(20 - x) = 4x - 40 = 0 \Rightarrow x = 10$. $\frac{d^2S}{dx^2} = 4 > 0$: minimum.
Parts: 10 and 10 (minimum sum of squares 200).

Similarly (Ex. 2.4 Q10): divide 30 into two parts with maximum product: $P = x(30 - x)$, $P' = 30 - 2x = 0 \Rightarrow x = 15$, $P'' = -2 < 0$; parts 15, 15.

Wire bent into a rectangle [PYQ Jun 2026 Q22, 3 marks; Ex. 2.4 Q12; textbook Ex. 2 with 120 cm]

A wire of length 36 cm is bent into a rectangle. Find its dimensions for maximum area.
Let length $= x$, breadth $= y$. Perimeter: $2(x + y) = 36 \Rightarrow y = 18 - x$.

$A = xy = x(18 - x) = 18x - x^2$. $\dfrac{dA}{dx} = 18 - 2x = 0 \Rightarrow x = 9$. $\dfrac{d^2A}{dx^2} = -2 < 0$: maximum.
$y = 18 - 9 = 9$. The area is maximum when the rectangle is a square of side 9 cm (area 81 cm²).

Open box with square base [PYQ Mar 2024 Q32, 4 marks]

A box with a square base and open top has surface area 147 cm². Find its dimensions for maximum volume.
Let the side of the base be x cm and height h cm.

Constraint (base + 4 sides, no top): $x^2 + 4xh = 147 \Rightarrow h = \dfrac{147 - x^2}{4x}$.

Objective: $V = x^2h = x^2 \cdot \dfrac{147 - x^2}{4x} = \dfrac{147x - x^3}{4}$.

$\dfrac{dV}{dx} = \dfrac{147 - 3x^2}{4} = 0 \Rightarrow x^2 = 49 \Rightarrow x = 7$ (reject $x = -7$).

$\dfrac{d^2V}{dx^2} = -\dfrac{6x}{4} = -\dfrac{3x}{2}$; at $x = 7$ it is $-\dfrac{21}{2} < 0$: maximum.

$h = \dfrac{147 - 49}{28} = \dfrac{98}{28} = 3.5$.

Dimensions: base 7 cm × 7 cm, height 3.5 cm. (Maximum volume $= 49 \times 3.5 = 171.5$ cm³.)

Ex. 2.4 Q17 is the same with 192 cm²: $x = 8$, $h = 4$. Notice $h = \frac{x}{2}$ in both — for an open square-based box of maximum volume, height is always half the base side. That is a good check of your arithmetic.

Box from a square sheet by cutting corners (textbook Ex. 4)

Squares of side x are cut from the corners of an 18 cm square card and the sides turned up.
$V = (18 - 2x)^2 x = 4x^3 - 72x^2 + 324x$. $V' = 12x^2 - 144x + 324 = 12(x - 3)(x - 9) = 0 \Rightarrow x = 3$ or 9.
$x = 9$ is rejected: the base side $18 - 2(9) = 0$. $V'' = 24x - 144$, $V''(3) = -72 < 0$: maximum.
Maximum volume $= (12)^2(3) = 432$ cm³.

Printed area on a sheet (textbook Ex. 3)

Sheet area 24 m²; margins 75 cm top and bottom, 50 cm each side. Let width x, length y; $xy = 24$.
Printed area $A = (x - 1)(y - 1.5) = (x - 1)\left(\dfrac{24}{x} - 1.5\right) = 25.5 - 1.5x - \dfrac{24}{x}$.

$A' = -1.5 + \dfrac{24}{x^2} = 0 \Rightarrow x^2 = 16 \Rightarrow x = 4$; $A'' = -\dfrac{48}{x^3} < 0$: maximum. $y = 6$.
Paper: width 4 m, length 6 m.

Textbook misprint

The textbook writes "$-1.5x + \frac{24}{x^2} = 0$" for $A' = 0$; the stray x is a typo. It should be $-1.5 + \frac{24}{x^2} = 0$, which is what leads to $x = 4$.

Triangle with two given sides (textbook Ex. 5)

$\Delta = \frac{1}{2}bc \sin A$. $\dfrac{d\Delta}{dA} = \frac{1}{2}bc \cos A = 0 \Rightarrow A = \dfrac{\pi}{2}$; $\dfrac{d^2\Delta}{dA^2} = -\frac{1}{2}bc \sin A < 0$. Maximum area when the angle between the given sides is $\dfrac{\pi}{2}$.

Cone of given slant height (textbook Ex. 6)

Slant l; height x, radius r: $r^2 = l^2 - x^2$. $V = \frac{1}{3}\pi r^2 x = \frac{\pi}{3}(l^2 x - x^3)$.

$V' = \frac{\pi}{3}(l^2 - 3x^2) = 0 \Rightarrow x = \dfrac{l}{\sqrt{3}}$; $V'' = -2\pi x < 0$: maximum. Then $r = \sqrt{l^2 - \frac{l^2}{3}} = \dfrac{\sqrt{2}\,l}{\sqrt{3}}$.

Semi-vertical angle α: $\tan\alpha = \dfrac{r}{x} = \sqrt{2} \Rightarrow \alpha = \tan^{-1}\sqrt{2}$.

Covered box, base $x \times 3x$, fixed volume — minimum surface (textbook Ex. 7)

$V = 3x^2 h$ (constant), $S = 2(3x^2) + 2(3xh) + 2(xh) = 6x^2 + 8xh$.

Since V is constant, $\dfrac{dV}{dx} = 0$: $3x^2\dfrac{dh}{dx} + 6xh = 0 \Rightarrow \dfrac{dh}{dx} = -\dfrac{2h}{x}$.

$\dfrac{dS}{dx} = 12x + 8\left(x\dfrac{dh}{dx} + h\right) = 12x + 8(-2h + h) = 12x - 8h = 0 \Rightarrow h = \dfrac{3x}{2}$.

$\dfrac{d^2S}{dx^2} = 12 - 8\dfrac{dh}{dx} = 12 + \dfrac{16h}{x} > 0$: minimum. Height $= \frac{3}{2}\times$ shorter side of base.

This example teaches a technique: when the constraint is awkward to solve, differentiate it **implicitly** instead.

Exercise answers (Ex. 2.4 Q13–Q22, Misc. II Q12–Q20)

Problem	Key equation	Answer
Ball $h = 3 + 14t - 5t^2$	$h' = 14 - 10t = 0$	$t = 1.4$ s, max height 12.8
Rectangle in semicircle, radius 1	vertices $(\pm\cos\theta, \sin\theta)$, $A = \sin 2\theta$	$\theta = \frac{\pi}{4}$: $\sqrt{2} \times \frac{1}{\sqrt{2}}$, area 1
Open cylinder, volume πa^3	$S = \pi r^2 + \frac{2\pi a^3}{r}$	$r = a, h = a$
Triangle perimeter 10, one side 4	$A^2 = 5(5-x)(x-1)$	other sides 3, 3
Profit $P = (150 - x)x - 1625$	$P' = 150 - 2x$	$x = 75$, $P_{\max} = 4000$
$x + y = 15$, maximise x^2y^3	$x(15-x)^2(30-5x) = 0$	$x = 6, y = 9$
Square has least perimeter for given area	$P = 2\left(x + \frac{A}{x}\right)$	$x = \sqrt{A}$
Closed cylinder, given volume, least surface	$S = 2\pi r^2 + \frac{2V}{r}$	$h = 2r$
Largest cylinder in sphere of radius r	$V = \pi\left(r^2 - \frac{h^2}{4}\right)h$	$h = \frac{2r}{\sqrt{3}}$, $V = \frac{4\pi r^3}{3\sqrt{3}}$
Open box from sheet of area a^2	$V = \frac{x}{4}(a^2 - x^2)$	$V_{\max} = \frac{a^3}{6\sqrt{3}}$
Window: rectangle + semicircle, perimeter 30	$A = 30r - 2r^2 - \frac{\pi r^2}{2}$	$r = \frac{30}{\pi+4}$; rectangle $\frac{60}{\pi+4} \times \frac{30}{\pi+4}$ m
Cone of max volume in sphere radius r	$V = \frac{\pi}{3}(2rh - h^2)h$	altitude $\frac{4r}{3}$
Cylinder in cone of height H	$V \propto (H-h)^2 h$	$h = \frac{H}{3}$
Sheet 8 : 15, removed squares total 100	$6x^2 - 23x + 15 = 0$	sides 24 and 45

How 4-mark optimisation answers are usually marked (typical scheme, not an official one)

Your project folder has the question papers but no official marking schemes, so this split is the usual pattern rather than a verified one: about **1 mark** for forming the objective in one variable, **1 mark** for differentiating and solving $f' = 0$, **1 mark** for the second-derivative test, **1 mark** for the final dimensions/value. Skipping the second-derivative test is the most common way to drop a mark.

10 Chapter summary

Everything on one page

Topic	Key result
Tangent at (x_1, y_1)	$y - y_1 = m(x - x_1)$, $m = \left(\frac{dy}{dx}\right)_{(x_1,y_1)}$
Normal	$y - y_1 = -\frac{1}{m}(x - x_1)$
Parallel / perpendicular	$m_1 = m_2$ / $m_1 m_2 = -1$; slope of $ax + by + c = 0$ is $-\frac{a}{b}$
Related rates	differentiate the linking formula w.r.t. t; substitute after
Motion	$v = \frac{ds}{dt}$, $a = \frac{d^2s}{dt^2}$, $j = \frac{d^3s}{dt^3}$
Approximation	$f(a+h) \approx f(a) + hf'(a)$; h in radians for trig
Rolle	cont. on $[a,b]$, diff. on (a,b), $f(a) = f(b) \Rightarrow \exists c \in (a,b) : f'(c) = 0$
LMVT	cont. on $[a,b]$, diff. on $(a,b) \Rightarrow \exists c : f'(c) = \frac{f(b)-f(a)}{b-a}$
Increasing / decreasing	$f' > 0$ / $f' < 0$
Max / min	$f'(c) = 0$ and $f''(c) < 0$ / $f''(c) > 0$; if $f''(c) = 0$ use sign change of f'

11 Self-assessment

Attempt without looking back. Answers follow.

1. Find the slope of the normal to $y = x^3 - x$ at $x = 2$.
2. Find the equation of the tangent to $y = e^x$ at the point where it crosses the Y-axis.
3. Find the approximate value of $\sqrt{25.2}$.
4. The radius of a circle increases at 0.5 cm/s. Find the rate of increase of its circumference.
5. Verify Rolle's theorem for $f(x) = x^2 - 6x + 8$ on $[2, 4]$.
6. Find c of LMVT for $f(x) = x^3$ on $[0, 3]$.
7. Find the intervals in which $f(x) = x^2 - 4x + 5$ is increasing.
8. Find the local minimum value of $f(x) = x^2 - 8x + 20$.
9. A rectangle has area 64 cm². Find the dimensions for least perimeter.
10. $s = t^3 - 6t^2 + 9t$. When is the particle at rest?
11. Find a and b if $y = x^2 + ax + b$ touches $y = x$ at $(1, 1)$.
12. Using LMVT idea: a car covers 150 km in 2 hours. What can you say about its speedometer

reading?

Answers to self-assessment

1. $y' = 3x^2 - 1 = 11$ at $x = 2$; normal slope $-\frac{1}{11}$.
2. Point $(0, 1)$, slope $e^0 = 1$: $y = x + 1$, i.e. $x - y + 1 = 0$.
3. $a = 25$, $h = 0.2$: $5 + \frac{0.2}{10} = 5.02$.
4. $C = 2\pi r$, $\frac{dC}{dt} = 2\pi(0.5) = \pi$ cm/s.
5. Polynomial; $f(2) = 0 = f(4)$; $f'(c) = 2c - 6 = 0 \Rightarrow c = 3 \in (2, 4)$. Verified.
6. $\frac{27-0}{3} = 9 = 3c^2 \Rightarrow c = \sqrt{3} \in (0, 3)$.
7. $f' = 2x - 4 > 0 \Rightarrow x > 2$.
8. $f' = 2x - 8 = 0 \Rightarrow x = 4$, $f'' = 2 > 0$; minimum $f(4) = 4$.
9. Square, 8 cm $\times$ 8 cm.
10. $v = 3t^2 - 12t + 9 = 3(t - 1)(t - 3) = 0 \Rightarrow t = 1$ s and $t = 3$ s.
11. Slope: $2(1) + a = 1 \Rightarrow a = -1$; point: $1 - 1 + b = 1 \Rightarrow b = 1$.
12. Average speed 75 km/h, so by LMVT at some instant the speedometer read exactly 75 km/h.